

## 10 Extensions of complete DVRs

Recall that in our  $AKLB$  setup,  $A$  is a Dedekind domain with fraction field  $K$ , the field  $L$  is a finite separable extension of  $K$ , and  $B$  is the integral closure of  $A$  in  $L$ ; as we proved in Theorem 5.33, this implies that  $B$  is also a Dedekind domain (with  $L$  as its fraction field), and we proved in Theorem 9.22 that if  $A$  is a complete DVR then  $B$  is a DVR. We now want to show that in this situation  $B$  is also complete.

**Definition 10.1.** Let  $K$  be a field with absolute value  $|\cdot|$  and let  $V$  be a  $K$ -vector space. A *norm* on  $V$  is a function  $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$  such that

- $\|v\| = 0$  if and only if  $v = 0$ .
- $\|\lambda v\| = |\lambda| \|v\|$  for all  $\lambda \in K$  and  $v \in V$ .
- $\|v + w\| \leq \|v\| + \|w\|$  for all  $v, w \in V$ .

Each norm  $\|\cdot\|$  induces a topology on  $V$  via the distance metric  $d(v, w) := \|v - w\|$ .

**Example 10.2.** Let  $V$  be a  $K$ -vector space with basis  $(e_i)$ , and for  $v \in V$  let  $v_i \in K$  denote the coefficient of  $e_i$  in  $v = \sum_i v_i e_i$ . The *sup-norm*  $\|v\|_\infty := \sup\{|v_i|\}$  is a norm on  $V$  (thus every vector space has at least one norm). If  $V$  is also a  $K$ -algebra, an absolute value  $|\cdot|$  on  $V$  (as a ring) is a norm on  $V$  (as a  $K$ -vector space) if and only if it extends the absolute value on  $K$  (fix  $v \neq 0$  and note that  $\|\lambda\| \|v\| = \|\lambda v\| = |\lambda| \|v\| \Leftrightarrow \|\lambda\| = |\lambda|$ ).

**Proposition 10.3.** Let  $V$  be a vector space of finite dimension over a complete field  $K$ . Every norm on  $V$  induces the same topology, in which  $V$  is a complete metric space.

*Proof.* See Problem Set 5. □

**Theorem 10.4.** Let  $A$  be a complete DVR with fraction field  $K$ , maximal ideal  $\mathfrak{p}$ , discrete valuation  $v_{\mathfrak{p}}$ , and absolute value  $|x|_{\mathfrak{p}} := c^{v_{\mathfrak{p}}(x)}$ , with  $0 < c < 1$ . Let  $L/K$  be a finite extension of degree  $n$ . The following hold.

- There is a unique absolute value  $|x| := |N_{L/K}(x)|_{\mathfrak{p}}^{1/n}$  on  $L$  that extends  $|\cdot|_{\mathfrak{p}}$ ;
- The field  $L$  is complete with respect to  $|\cdot|$ , and its valuation ring  $\{x \in L : |x| \leq 1\}$  is equal to the integral closure  $B$  of  $A$  in  $L$ ;
- If  $L/K$  is separable then  $B$  is a complete DVR whose maximal ideal  $\mathfrak{q}$  induces

$$|x| = |x|_{\mathfrak{q}} := c^{\frac{1}{e_{\mathfrak{q}}} v_{\mathfrak{q}}(x)},$$

where  $e_{\mathfrak{q}}$  is the ramification index of  $\mathfrak{q}$ , that is,  $\mathfrak{p}B = \mathfrak{q}^{e_{\mathfrak{q}}}$ .

*Proof.* Assuming for the moment that  $|\cdot|$  is actually an absolute value (which is not obvious!), for any  $x \in K$  we have

$$|x| = |N_{L/K}(x)|_{\mathfrak{p}}^{1/n} = |x^n|_{\mathfrak{p}}^{1/n} = |x|_{\mathfrak{p}},$$

so  $|\cdot|$  extends  $|\cdot|_{\mathfrak{p}}$  and is therefore a norm on  $L$ . The fact that  $|\cdot|_{\mathfrak{p}}$  is nontrivial means that  $|x|_{\mathfrak{p}} \neq 1$  for some  $x \in K^\times$ , and  $|x|^a = |x|_{\mathfrak{p}} = |x|$  only for  $a = 1$ , which implies that  $|\cdot|$  is the unique absolute value in its equivalence class extending  $|\cdot|_{\mathfrak{p}}$ . Every norm on  $L$  induces the same topology (by Proposition 10.3), so  $|\cdot|$  is the only absolute value on  $L$  that extends  $|\cdot|_{\mathfrak{p}}$ .

We now show  $|\cdot|$  is an absolute value. Clearly  $|x| = 0 \Leftrightarrow x = 0$  and  $|\cdot|$  is multiplicative; we only need to check the triangle inequality. It suffices to show  $|x| \leq 1 \Rightarrow |x + 1| \leq |x| + 1$ ,

since we always have  $|y + z| = |z||y/z + 1|$  and  $|y| + |z| = |z|(|y/z| + 1)$ , and without loss of generality we assume  $|y| \leq |z|$ . In fact the stronger implication  $|x| \leq 1 \Rightarrow |x + 1| \leq 1$  holds:

$$|x| \leq 1 \iff |N_{L/K}(x)|_{\mathfrak{p}} \leq 1 \iff N_{L/K}(x) \in A \iff x \in B \iff x + 1 \in B \iff |x + 1| \leq 1.$$

The first biconditional follows from the definition of  $|\cdot|$ , the second follows from the definition of  $|\cdot|_{\mathfrak{p}}$ , the third is Corollary 9.21, the fourth is obvious, and the fifth follows from the first three after replacing  $x$  with  $x + 1$ . This completes the proof of (i), and also proves (ii).

We now assume  $L/K$  is separable. Then  $B$  is a DVR, by Theorem 9.22, and it is complete because it is the valuation ring of  $L$ . Let  $\mathfrak{q}$  be the unique maximal ideal of  $B$ . The valuation  $v_{\mathfrak{q}}$  extends  $v_{\mathfrak{p}}$  with index  $e_{\mathfrak{q}}$ , by Theorem 8.20, so  $v_{\mathfrak{q}}(x) = e_{\mathfrak{q}}v_{\mathfrak{p}}(x)$  for  $x \in K^{\times}$ . We have  $0 < c^{1/e_{\mathfrak{q}}} < 1$ , so  $|x|_{\mathfrak{q}} := (c^{1/e_{\mathfrak{q}}})^{v_{\mathfrak{q}}(x)}$  is an absolute value on  $L$  induced by  $v_{\mathfrak{q}}$ . To show it is equal to  $|\cdot|$ , it suffices to show that it extends  $|\cdot|_{\mathfrak{p}}$ , since we already know that  $|\cdot|$  is the unique absolute value on  $L$  with this property. For  $x \in K^{\times}$  we have

$$|x|_{\mathfrak{q}} = c^{\frac{1}{e_{\mathfrak{q}}}v_{\mathfrak{q}}(x)} = c^{\frac{1}{e_{\mathfrak{q}}}e_{\mathfrak{q}}v_{\mathfrak{p}}(x)} = c^{v_{\mathfrak{p}}(x)} = |x|_{\mathfrak{p}},$$

and the theorem follows.  $\square$

**Remark 10.5.** The transitivity of  $N_{L/K}$  in towers (Corollary 5.8) implies that we can uniquely extend the absolute value on the fraction field  $K$  of a complete DVR to an algebraic closure  $\overline{K}$ . In fact, this is another form of Hensel's lemma in the following sense: one can show that a (not necessarily discrete) valuation ring  $A$  is Henselian if and only if the absolute value of its fraction field  $K$  can be uniquely extended to  $\overline{K}$ ; see [4, Theorem 6.6].

**Corollary 10.6.** Assume  $AKLB$  and that  $A$  is a complete DVR with maximal ideal  $\mathfrak{p}$  and let  $\mathfrak{q}|\mathfrak{p}$ . Then  $v_{\mathfrak{q}}(x) = \frac{1}{f_{\mathfrak{q}}}v_{\mathfrak{p}}(N_{L/K}(x))$  for all  $x \in L$ .

*Proof.*  $v_{\mathfrak{p}}(N_{L/K}(x)) = v_{\mathfrak{p}}(N_{L/K}((x))) = v_{\mathfrak{p}}(N_{L/K}(\mathfrak{q}^{v_{\mathfrak{q}}(x)})) = v_{\mathfrak{p}}(\mathfrak{p}^{f_{\mathfrak{q}}v_{\mathfrak{q}}(x)}) = f_{\mathfrak{q}}v_{\mathfrak{q}}(x)$ .  $\square$

**Remark 10.7.** One can generalize the notion of a discrete valuation to a *valuation*, a surjective homomorphism  $v: K^{\times} \rightarrow \Gamma$ , in which  $\Gamma$  is a (totally) ordered abelian group and  $v(x + y) \geq \min(v(x), v(y))$ ; we extend  $v$  to  $K$  by defining  $v(0) = \infty$  to be strictly greater than any element of  $\Gamma$ . In the  $AKLB$  setup with  $A$  a complete DVR, one can then define a valuation  $v(x) = \frac{1}{e_{\mathfrak{q}}}v_{\mathfrak{q}}(x)$  with image  $\frac{1}{e_{\mathfrak{q}}}\mathbb{Z}$  that restricts to the discrete valuation  $v_{\mathfrak{p}}$  on  $K$ . The valuation  $v$  then extends to a valuation on  $\overline{K}$  with  $\Gamma = \mathbb{Q}$ . Some texts take this approach, but we will generally stick with discrete valuations (so our absolute value on  $L$  restricts to  $K$ , but our discrete valuations on  $L$  do not restrict to discrete valuations on  $K$ , they extend them with index  $e_{\mathfrak{q}}$ ).

**Remark 10.8.** Recall that a *valuation ring* is an integral domain  $A$  with fraction field  $K$  such that for every  $x \in K^{\times}$  either  $x \in A$  or  $x^{-1} \in A$  (possibly both). As you will show on Problem Set 6, if  $A$  is a valuation ring, then there exists a valuation  $v: K \rightarrow \Gamma \cup \{\infty\}$  for some totally ordered abelian group  $\Gamma$  such that  $A = \{x \in K : v(x) \geq 0\}$  is the valuation ring of  $K$  with respect to this valuation.

## 10.1 The Dedekind-Kummer theorem in a local setting

Recall that the Dedekind-Kummer theorem (Theorem 6.14) allows us to factor primes in our  $AKLB$  setting by factoring polynomials over the residue field, provided that  $B$  is monogenic

(of the form  $A[\alpha]$  for some  $\alpha \in B$ ), or the prime of interest does not contain the conductor. We now show that in the special case where  $A$  and  $B$  are DVRs and the residue field extension is separable,  $B$  is always monogenic; this holds, for example, whenever  $K$  is a local field. To prove this, we first recall a form of Nakayama's lemma.

**Lemma 10.9** (NAKAYAMA'S LEMMA). *Let  $A$  be a local ring with maximal ideal  $\mathfrak{p}$ , and let  $M$  be a finitely generated  $A$ -module. If the images of  $x_1, \dots, x_n \in M$  generate  $M/\mathfrak{p}M$  as an  $(A/\mathfrak{p})$ -vector space then  $x_1, \dots, x_n$  generate  $M$  as an  $A$ -module.*

*Proof.* See [1, Corollary 4.8b].  $\square$

Before proving our theorem on local monogenicity, let us record some corollaries of Nakayama's Lemma that will be useful to us later.

**Corollary 10.10.** *Let  $A$  be a local noetherian ring with maximal ideal  $\mathfrak{p}$ , let  $g \in A[x]$  be monic, and let  $B := A[x]/(g(x))$ . Every maximal ideal  $\mathfrak{m}$  of  $B$  contains the ideal  $\mathfrak{p}B$ .*

*Proof.* Suppose not. Then  $\mathfrak{m} + \mathfrak{p}B = B$  for some maximal ideal  $\mathfrak{m}$  of  $B$ . The ring  $B$  is finitely generated over the noetherian ring  $A$ , hence a noetherian  $A$ -module, so its  $A$ -submodules are all finitely generated. Let  $z_1, \dots, z_n$  be  $A$ -module generators for  $\mathfrak{m}$ . Every coset of  $\mathfrak{p}B$  in  $B$  can be written as  $z + \mathfrak{p}B$  for some  $A$ -linear combination  $z$  of  $z_1, \dots, z_n$ , so the images of  $z_1, \dots, z_n$  generate  $B/\mathfrak{p}B$  as an  $(A/\mathfrak{p})$ -vector space. By Nakayama's lemma,  $z_1, \dots, z_n$  generate  $B$ , in which case  $\mathfrak{m} = B$ , a contradiction.  $\square$

As a corollary, we immediately obtain a local version of the Dedekind-Kummer theorem that does not require  $A$  and  $B$  to be Dedekind domains.

**Corollary 10.11.** *Let  $A$  be a local noetherian ring with maximal ideal  $\mathfrak{p}$ , let  $g \in A[x]$  be a monic polynomial with reduction  $\bar{g} \in (A/\mathfrak{p})[x]$ , and let  $\alpha$  be the image of  $x$  in the ring  $B := A[x]/(g(x)) = A[\alpha]$ . The maximal ideals of  $B$  are  $(\mathfrak{p}, g_i(\alpha))$ , where  $g_1, \dots, g_m \in A[x]$  are lifts of the distinct irreducible polynomials  $\bar{g}_i \in (A/\mathfrak{p})[x]$  that divide  $\bar{g}$ .*

*Proof.* By Corollary 10.10, the quotient map  $B \rightarrow B/\mathfrak{p}B$  gives a one-to-one correspondence between maximal ideals of  $B$  and maximal ideals of  $B/\mathfrak{p}B$ , and we have

$$\frac{B}{\mathfrak{p}B} \simeq \frac{A[x]}{(\mathfrak{p}, g(x))} \simeq \frac{(A/\mathfrak{p})[x]}{(\bar{g}(x))}.$$

Each maximal ideal of  $(A/\mathfrak{p})[x]/(\bar{g}(x))$  is the reduction of an irreducible divisor of  $\bar{g}$ , hence one of the  $\bar{g}_i$  (because  $(A/\mathfrak{p})[x]$  is a PID). The corollary follows.  $\square$

**Theorem 10.12.** *Assume  $AKLB$ , with  $A$  and  $B$  DVRs with residue fields  $k := A/\mathfrak{p}$  and  $l := B/\mathfrak{q}$ . If  $l/k$  is separable then  $B = A[\alpha]$  for some  $\alpha \in B$ ; if  $L/K$  is unramified this holds for every lift  $\alpha$  of any generator  $\bar{\alpha}$  for  $l = k(\bar{\alpha})$ .*

*Proof.* Let  $\mathfrak{p}B = \mathfrak{q}^e$  be the factorization of  $\mathfrak{p}B$  and let  $f = [l : k]$  be the residue field degree, so that  $ef = n := [L : K]$ . The extension  $l/k$  is separable, so we may apply the primitive element theorem to write  $l = k(\bar{\alpha}_0)$  for some  $\bar{\alpha}_0 \in l$  whose minimal polynomial  $\bar{g}$  is separable of degree equal to  $f$ . Let  $g \in A[x]$  be a monic lift of  $\bar{g}$ , and let  $\alpha_0$  be any lift of  $\bar{\alpha}_0$  to  $B$ . If  $v_{\mathfrak{q}}(g(\alpha_0)) = 1$  then let  $\alpha := \alpha_0$ . Otherwise, let  $\pi_0$  be any uniformizer for  $B$  and let  $\alpha := \alpha_0 + \pi_0 \in B$  (so  $\alpha \equiv \bar{\alpha}_0 \pmod{\mathfrak{q}}$ ), and writing  $g(x + \pi_0) = g(x) + \pi_0 g'(x) + \pi_0^2 h(x)$  for some  $h \in A[x]$  via Lemma 9.11, we have

$$v_{\mathfrak{q}}(g(\alpha)) = v_{\mathfrak{q}}(g(\alpha_0 + \pi_0)) = v_{\mathfrak{q}}(g(\alpha_0) + \pi_0 g'(\alpha_0) + \pi_0^2 h(\alpha_0)) = 1,$$

so  $\pi := g(\alpha)$  is also a uniformizer for  $B$ .

We now claim  $B = A[\alpha]$ , equivalently, that  $1, \alpha, \dots, \alpha^{n-1}$  generate  $B$  as an  $A$ -module. By Nakayama's lemma, it suffices to show that the reductions of  $1, \alpha, \dots, \alpha^{n-1}$  span  $B/\mathfrak{p}B$  as a  $k$ -vector space. We have  $\mathfrak{p}B = \mathfrak{q}^e$ , so  $\mathfrak{p}B = (\pi^e)$ . We can represent each element of  $B/\mathfrak{p}B$  as a coset

$$b + \mathfrak{p}B = b_0 + b_1\pi + b_2\pi^2 + \dots + b_{e-1}\pi^{e-1} + \mathfrak{p}B,$$

where  $b_0, \dots, b_{e-1}$  are determined up to equivalence modulo  $\pi B$ . Now  $1, \bar{\alpha}, \dots, \bar{\alpha}^{f-1}$  are a basis for  $B/\pi B = B/\mathfrak{q}$  as a  $k$ -vector space, and  $\pi = g(\alpha)$ , so we can rewrite this as

$$\begin{aligned} b + \mathfrak{p}B &= (a_0 + a_1\alpha + \dots + a_{f-1}\alpha^{f-1}) \\ &\quad + (a_f + a_{f+1}\alpha + \dots + a_{2f-1}\alpha^{f-1})g(\alpha) \\ &\quad + \dots \\ &\quad + (a_{ef-f+1} + a_{ef-f+2}\alpha + \dots + a_{ef-1}\alpha^{f-1})g(\alpha)^{e-1} + \mathfrak{p}B. \end{aligned}$$

Since  $\deg g = f$ , and  $n = ef$ , this expresses  $b + \mathfrak{p}B$  in the form  $b' + \mathfrak{p}B$  with  $b'$  in the  $A$ -span of  $1, \dots, \alpha^{n-1}$ . Thus  $B = A[\alpha]$ .

We now note that if  $L/K$  is unramified then  $l/k$  is separable (this is part of the definition of unramified), and  $e = 1$ ,  $f = n$ , in which case there is no need to require  $g(\alpha)$  to be a uniformizer and we can just take  $\alpha = \alpha_0$  to be any lift of any  $\bar{\alpha}_0$  that generates  $l$  over  $k$ .  $\square$

In our  $AKLB$  setup, if  $A$  is a complete DVR with maximal ideal  $\mathfrak{p}$  then  $B$  is a complete DVR with maximal ideal  $\mathfrak{q}|\mathfrak{p}$  and the formula  $[L : K] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}}f_{\mathfrak{q}}$  given by Theorem 5.43 has only one term  $e_{\mathfrak{q}}f_{\mathfrak{q}}$ . We now simplify matters even further by reducing to the two extreme cases  $f_{\mathfrak{q}} = 1$  (a totally ramified extension) and  $e_{\mathfrak{q}} = 1$  (an unramified extension, provided that the residue field extension is separable).<sup>1</sup>

## 10.2 Unramified extensions of a complete DVR

Let  $A$  be a complete DVR with fraction field  $K$  and residue field  $k$ . Associated to any finite unramified extension of  $L/K$  of degree  $n$  is a corresponding finite separable extension of residue fields  $l/k$  of the same degree  $n$ . Given that the extensions  $L/K$  and  $l/k$  are finite separable extensions of the same degree, we might wonder how they are related. More precisely, if we fix  $K$  with residue field  $k$ , what is the relationship between finite unramified extensions  $L/K$  of degree  $n$  and finite separable extensions  $l/k$  of degree  $n$ ? Each  $L/K$  uniquely determines a corresponding  $l/k$ , but what about the converse?

This question has a surprisingly nice answer. The finite unramified extensions  $L$  of  $K$  form a category  $\mathcal{C}_K^{\text{unr}}$  whose morphisms are  $K$ -algebra homomorphisms, and the finite separable extensions  $l$  of  $k$  form a category  $\mathcal{C}_k^{\text{sep}}$  whose morphisms are  $k$ -algebra homomorphisms. These two categories are equivalent.

**Theorem 10.13.** *Let  $A$  be a complete DVR with fraction field  $K$  and residue field  $k := A/\mathfrak{p}$ . The categories  $\mathcal{C}_K^{\text{unr}}$  and  $\mathcal{C}_k^{\text{sep}}$  are equivalent via the functor  $\mathcal{F}: \mathcal{C}_K^{\text{unr}} \rightarrow \mathcal{C}_k^{\text{sep}}$  that sends each unramified extension  $L$  of  $K$  to its residue field  $l$ , and each  $K$ -algebra homomorphism  $\varphi: L_1 \rightarrow L_2$  to the  $k$ -algebra homomorphism  $\bar{\varphi}: l_1 \rightarrow l_2$  defined by  $\bar{\varphi}(\bar{\alpha}) := \varphi(\alpha)$ , where  $\alpha$*

<sup>1</sup>Recall from Definition 5.45 that separability of the residue field extension is part of the definition of an unramified extension. If the residue field is perfect (as when  $K$  is a local field, for example), the residue field extension is automatically separable, but in general it need not be, even when  $L/K$  is unramified.

is any lift of  $\bar{\alpha} \in l_1 := B_1/\mathfrak{q}_1$  to  $B_1$  and  $\overline{\varphi(\alpha)}$  is the reduction of  $\varphi(\alpha) \in B_2$  to  $l_2 := B_2/\mathfrak{q}_2$ ; here  $\mathfrak{q}_1, \mathfrak{q}_2$  are the maximal ideals of the valuation rings  $B_1, B_2$  of  $L_1, L_2$ , respectively.

In particular,  $\mathcal{F}$  gives a bijection between the isomorphism classes in  $\mathcal{C}_K^{\text{unr}}$  and  $\mathcal{C}_k^{\text{sep}}$ , and if  $L_1, L_2$  have residue fields  $l_1, l_2$  then  $\mathcal{F}$  induces a bijection of finite sets

$$\text{Hom}_K(L_1, L_2) \xrightarrow{\sim} \text{Hom}_k(l_1, l_2).$$

*Proof.* Let us first verify that  $\mathcal{F}$  is well-defined. It is clear that it maps finite unramified extensions  $L/K$  to finite separable extensions  $l/k$ , but we should check that the map on morphisms does not depend on the lift  $\alpha$  of  $\bar{\alpha}$  we pick. So let  $\varphi: L_1 \rightarrow L_2$  be a  $K$ -algebra homomorphism, and for  $\bar{\alpha} \in l_1$ , let  $\alpha$  and  $\alpha'$  be two lifts of  $\bar{\alpha}$  to  $B_1$ . Then  $\alpha - \alpha' \in \mathfrak{q}_1$ , and this implies that  $\varphi(\alpha - \alpha') \in \varphi(\mathfrak{q}_1) = \varphi(B_1) \cap \mathfrak{q}_2 \subseteq \mathfrak{q}_2$ , and therefore  $\overline{\varphi(\alpha)} = \overline{\varphi(\alpha')}$ . The identity  $\varphi(\mathfrak{q}_1) = \varphi(B_1) \cap \mathfrak{q}_2 \subseteq \mathfrak{q}_2$  follows from the fact that  $\varphi$  restricts to an injective ring homomorphism  $B_1 \rightarrow B_2$  and  $B_2/\varphi(B_1)$  is a finite extension of DVRs in which  $\mathfrak{q}_2$  lies over the prime  $\varphi(\mathfrak{q}_1)$  of  $\varphi(B_1)$ . It's easy to see that  $\mathcal{F}$  sends identity morphisms to identity morphisms and that it is compatible with composition, so we have a well-defined functor.

To show that  $\mathcal{F}$  is an equivalence of categories we need to prove two things:

- $\mathcal{F}$  is essentially surjective: each separable  $l/k$  is isomorphic to the residue field of some unramified  $L/K$
- $\mathcal{F}$  is full and faithful: the induced map  $\text{Hom}_K(L_1, L_2) \rightarrow \text{Hom}_k(l_1, l_2)$  is a bijection.

We first show that  $\mathcal{F}$  is essentially surjective. Given a finite separable extension  $l/k$ , we may apply the primitive element theorem to write

$$l \simeq k(\bar{\alpha}) = \frac{k[x]}{(\bar{g}(x))},$$

for some  $\bar{\alpha} \in l$  whose minimal polynomial  $\bar{g} \in k[x]$  is necessarily monic, irreducible, separable, and of degree  $n := [l : k]$ . Let  $g \in A[x]$  be any monic lift of  $\bar{g}$ ; then  $g$  is also irreducible, separable, and of degree  $n$ . Now let

$$L := \frac{K[x]}{(g(x))} = K(\alpha),$$

where  $\alpha$  is the image of  $x$  in  $K[x]/g(x)$ . Then  $L/K$  is a finite separable extension, and by Corollary 10.11,  $(\mathfrak{p}, g(\alpha)) = (\mathfrak{p}, 0) = \mathfrak{p}A[\alpha]$  is the unique maximal ideal of  $A[\alpha]$ , since  $\bar{g}$  is irreducible, and

$$\frac{B}{\mathfrak{q}} \simeq \frac{A[\alpha]}{(\mathfrak{p}, g(\alpha))} \simeq \frac{A[x]}{(\mathfrak{p}, g(x))} \simeq \frac{(A/\mathfrak{p})[x]}{(\bar{g}(x))} \simeq l,$$

where  $B$  is the valuation ring of  $L$  with maximal ideal  $\mathfrak{q}$ . Thus  $[L : K] = \deg g = [l : k] = n$ , and it follows that  $L/K$  is an unramified extension of degree  $n = f := [l : k]$ : the ramification index of  $\mathfrak{q}$  is necessarily  $e = n/f = 1$ , and the extension  $l/k$  is separable by assumption (so in fact  $B = A[\alpha]$ , by Theorem 10.12).

We now show that the functor  $\mathcal{F}$  is full and faithful. Given finite unramified extensions  $L_1, L_2$  with valuation rings  $B_1, B_2$  and residue fields  $l_1, l_2$ , we have induced maps

$$\text{Hom}_K(L_1, L_2) \xrightarrow{\sim} \text{Hom}_A(B_1, B_2) \longrightarrow \text{Hom}_k(l_1, l_2).$$

The first map is given by restriction from  $L_1$  to  $B_1$ , and since tensoring with  $K$  gives an inverse map in the other direction, it is a bijection. We need to show that the same is

true of the second map, which sends  $\varphi: B_1 \rightarrow B_2$  to the  $k$ -homomorphism  $\bar{\varphi}$  that sends  $\bar{\alpha} \in l_1 = B_1/\mathfrak{q}_1$  to the reduction of  $\varphi(\alpha)$  modulo  $\mathfrak{q}_2$ , where  $\alpha$  is any lift of  $\bar{\alpha}$ .

As above, use the primitive element theorem to write  $l_1 = k(\bar{\alpha}) = k[x]/(\bar{g}(x))$  for some  $\bar{\alpha} \in l_1$ . If we now lift  $\bar{\alpha}$  to  $\alpha \in B_1$ , we must have  $L_1 = K(\alpha)$ , since  $[L_1 : K] = [l_1 : k]$  is equal to the degree of the minimal polynomial  $g$  of  $\alpha$  which cannot be less than the degree of the minimal polynomial  $\bar{g}$  of  $\bar{\alpha}$  (both are monic). Moreover, we also have  $B_1 = A[\alpha]$ , since this is true of the valuation ring of every finite unramified extension in our category.

Each  $A$ -algebra homomorphism in

$$\mathrm{Hom}_A(B_1, B_2) = \mathrm{Hom}_A\left(\frac{A[x]}{(\bar{g}(x))}, B_2\right)$$

is uniquely determined by the image of  $x$  in  $B_2$ . This gives a bijection between  $\mathrm{Hom}_A(B_1, B_2)$  and the roots of  $g$  in  $B_2$ . Similarly, each  $k$ -algebra homomorphism in

$$\mathrm{Hom}_k(l_1, l_2) = \mathrm{Hom}_k\left(\frac{k[x]}{(\bar{g}(x))}, l_2\right)$$

is uniquely determined by the image of  $x$  in  $l_2$ , and there is a bijection between  $\mathrm{Hom}_k(l_1, l_2)$  and the roots of  $\bar{g}$  in  $l_2$ . Now  $\bar{g}$  is separable, so every root of  $\bar{g}$  in  $l_2 = B_2/\mathfrak{q}_2$  lifts to a unique root of  $g$  in  $B_2$ , by Hensel's Lemma 9.15. Thus the map  $\mathrm{Hom}_A(B_1, B_2) \rightarrow \mathrm{Hom}_k(l_1, l_2)$  induced by  $\mathcal{F}$  is a bijection.  $\square$

**Remark 10.14.** In the proof above we actually only used the fact that  $L_1/K$  is unramified. The map  $\mathrm{Hom}_K(L_1, L_2) \rightarrow \mathrm{Hom}_k(l_1, l_2)$  is a bijection even if  $L_2/K$  is not unramified.

Let us note the following corollary, which follows from our proof of Theorem 10.13.

**Corollary 10.15.** *Assume  $AKLB$  with  $A$  a complete DVR with residue field  $k$ . Then  $L/K$  is unramified if and only if  $B = A[\alpha]$  for some  $\alpha \in L$  whose minimal polynomial  $g \in A[x]$  has separable image  $\bar{g}$  in  $k[x]$ .*

*Proof.* The forward direction was proved in the proof of the theorem, and for the reverse direction note that  $\bar{g}$  must be irreducible, since otherwise we could use Hensel's lemma to lift a non-trivial factorization of  $\bar{g}$  to a non-trivial factorization of  $g$ , so the residue field extension is separable and has the same degree as  $L/K$ , so  $L/K$  is unramified.  $\square$

**Corollary 10.16.** *Let  $A$  be a complete DVR with fraction field  $K$  and residue field  $k$ , and let  $\zeta_n$  be a primitive  $n$ th root of unity in some algebraic closure of  $K$ , with  $n$  prime to the characteristic of  $k$ . The extension  $K(\zeta_n)/K$  is unramified.*

*Proof.* The field  $K(\zeta_n)$  is the splitting field of  $f(x) = x^n - 1$  over  $K$ . The image  $\bar{f}$  of  $f$  in  $k[x]$  is separable when  $p \nmid n$ , since  $\gcd(\bar{f}, \bar{f}') \neq 1$  only when  $\bar{f}' = nx^{n-1}$  is zero, equivalently, only when  $p|n$ . When  $\bar{f}$  is separable, so are all of its divisors, including the reduction of the minimal polynomial of  $\zeta_n$ , which must be irreducible since otherwise we could obtain a contradiction by lifting a non-trivial factorization via Hensel's lemma. It follows that the residue field of  $K(\zeta_n)$  is a separable extension of  $k$ , thus  $K(\zeta_n)/K$  is unramified.  $\square$

When the residue field  $k$  is finite (always the case if  $K$  is a local field), we can give a precise description of the finite unramified extensions  $L/K$ .

**Corollary 10.17.** *Let  $A$  be a complete DVR with fraction field  $K$  and finite residue field  $\mathbb{F}_q$ , and let  $L$  be a degree  $n$  extension of  $K$ . Then  $L/K$  is unramified if and only if  $L \simeq K(\zeta_{q^n-1})$ . When this holds,  $A[\zeta_{q^n-1}]$  is the integral closure of  $A$  in  $L$  and  $L/K$  is a Galois extension with  $\text{Gal}(L/K) \simeq \mathbb{Z}/n\mathbb{Z}$ .*

*Proof.* The reverse implication is implied by Corollary 10.16; note that  $K(\zeta_{q^n-1})$  has degree  $n$  over  $K$  because its residue field is the splitting field of  $x^{q^n-1} - 1$  over  $\mathbb{F}_q$ , which is an extension of degree  $n$  (indeed, one can take this as the definition of  $\mathbb{F}_{q^n}$ ).

Suppose  $L/K$  is unramified. Then  $[l : k] = [L : K] = n$  and  $l \simeq \mathbb{F}_{q^n}$  has multiplicative group cyclic of order  $q^n - 1$  generated by some  $\bar{\alpha}$ . The minimal polynomial  $\bar{g} \in \mathbb{F}_q[x]$  of  $\bar{\alpha}$  divides  $x^{q^n-1} - 1$ , and since  $\bar{g}$  is irreducible, it is coprime to the quotient  $(x^{q^n-1} - 1)/\bar{g}$ . By Hensel's Lemma 9.19, we can lift  $\bar{g}$  to a polynomial  $g \in A[x]$  that divides  $x^{q^n-1} - 1 \in A[x]$ , and by Hensel's Lemma 9.15 we can lift  $\bar{\alpha}$  to a root  $\alpha$  of  $g$ , in which case  $\alpha$  is also a root of  $x^{q^n-1} - 1$ ; it must be a primitive  $(q^n - 1)$ -root of unity because its reduction  $\bar{\alpha}$  is.

Let  $B$  be the integral closure of  $A$  in  $L$ . We have  $B \simeq A[\zeta_{q^n-1}]$  by Theorem 10.12, and  $L$  is the splitting field of  $x^{q^n-1} - 1$ , since its residue field  $\mathbb{F}_{q^n}$  is (we can lift the factorization of  $x^{q^n-1} - 1$  from  $\mathbb{F}_{q^n}$  to  $L$  via Hensel's lemma). It follows that  $L/K$  is Galois, and the bijection between  $(q^n - 1)$ -roots of unity in  $L$  and  $\mathbb{F}_{q^n}$  induces an isomorphism  $\text{Gal}(L/K) \simeq \text{Gal}(l/k) = \text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) \simeq \mathbb{Z}/n\mathbb{Z}$ .  $\square$

**Example 10.18.** Consider  $A = \mathbb{Z}_p$ ,  $K = \mathbb{Q}_p$ ,  $k = \mathbb{F}_p$ , and fix  $\bar{\mathbb{F}}_p$  and  $\bar{\mathbb{Q}}_p$ . For each positive integer  $n$ , the finite field  $\mathbb{F}_p$  has a unique extension of degree  $n$  in  $\bar{\mathbb{F}}_p$ , namely,  $\mathbb{F}_{p^n}$ . Thus for each positive integer  $n$ , the local field  $\mathbb{Q}_p$  has a unique unramified extension of degree  $n$ ; it can be explicitly constructed by adjoining a primitive root of unity  $\zeta_{p^n-1}$  to  $\mathbb{Q}_p$ . The element  $\zeta_{p^n-1}$  will necessarily have minimal polynomial of degree  $n$  dividing  $x^{p^n-1} - 1$ .

Another useful consequence of Theorem 10.13 that applies when the residue field is finite is that the norm map  $N_{L/K}$  restricts to a surjective map  $B^\times \rightarrow A^\times$  on unit groups; in fact, this property characterizes unramified extensions.

**Theorem 10.19.** *Assume  $AKLB$  with  $A$  a complete DVR with finite residue field. Then  $L/K$  is unramified if and only if  $N_{L/K}(B^\times) = A^\times$ .*

*Proof.* See Problem Set 6.  $\square$

**Definition 10.20.** Let  $L/K$  be a separable extension. The *maximal unramified extension of  $K$  in  $L$*  is the subfield

$$\bigcup_{\substack{K \subseteq E \subseteq L \\ E/K \text{ fin. unram.}}} E \subseteq L$$

where the union is over finite unramified subextensions  $E/K$ . When  $L = K^{\text{sep}}$  is the separable closure of  $K$ , this is the *maximal unramified extension of  $K$* , denoted  $K^{\text{unr}}$ .

**Example 10.21.** The field  $\mathbb{Q}_p^{\text{unr}}$  is an infinite extension of  $\mathbb{Q}_p$  with Galois group

$$\text{Gal}(\mathbb{Q}_p^{\text{unr}}/\mathbb{Q}_p) \simeq \text{Gal}(\bar{\mathbb{F}}_p/\mathbb{F}_p) = \varprojlim_n \text{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) \simeq \varprojlim_n \mathbb{Z}/n\mathbb{Z} =: \hat{\mathbb{Z}},$$

where the inverse limit is taken over positive integers  $n$  ordered by divisibility. The ring  $\hat{\mathbb{Z}}$  is the *profinite completion* of  $\mathbb{Z}$ . The field  $\mathbb{Q}_p^{\text{unr}}$  has value group  $\mathbb{Z}$  and residue field  $\bar{\mathbb{F}}_p$ .

**Theorem 10.22.** *Assume  $AKLB$  with  $A$  a complete DVR and separable residue field extension  $l/k$ . Let  $e$  and  $f$  be the ramification index and residue field degrees, respectively, and let  $\mathfrak{q}$  be the unique prime of  $B$ . The following hold:*

- (i) *There is a unique intermediate field extension  $E/K$  that contains every unramified extension of  $K$  in  $L$  and it has degree  $[E : K] = f$ .*
- (ii) *The extension  $L/E$  is totally ramified and has degree  $[L : E] = e$ .*
- (iii) *If  $L/K$  is Galois then  $\text{Gal}(L/K)$  is the decomposition group of  $D_{\mathfrak{q}}$ ,  $\text{Gal}(L/E)$  is the inertia subgroup of  $I_{\mathfrak{q}}$ , and  $E/K$  is Galois with  $\text{Gal}(E/K) \simeq D_{\mathfrak{q}}/I_{\mathfrak{q}} \simeq \text{Gal}(l/k)$ .*

*Proof.* (i) Let  $E/K$  be the finite unramified extension of  $K$  in  $L$  corresponding to the finite separable extension  $l/k$  given by Theorem 10.13; then  $[E : K] = [l : k] = f$  as desired. The maximal unramified extension  $E'$  of  $K$  in  $L$  has the same residue field  $l$  as  $L$ , which is also the residue field of  $E$ , and equivalence of categories given by Theorem 10.13 implies that the trivial isomorphism  $\ell \simeq \ell$  corresponds to an isomorphism  $E \simeq E'$  that allows us to view  $E$  as a subfield of  $L$ ; the same applies to any unramified extension of  $K$  with residue field  $l$ , so  $E$  is unique up to isomorphism.

(ii) Let  $n = [L : K]$ . Then  $[L : E] = [L : K]/[E : K] = n/f = ef/f = e$ .

(iii) We have  $D_{\mathfrak{q}} \subseteq \text{Gal}(L/K)$  of order  $ef = [L : K]$ , so this inclusion is an equality. If we put  $\mathfrak{q}_E := \mathfrak{q} \cap E$  then Proposition 7.13 implies  $I_{\mathfrak{q}_E} = \text{Gal}(L/E) \cap I_{\mathfrak{q}}$ . These three groups all have order  $e$  and must coincide. The group  $I_{\mathfrak{q}}$  is normal in  $D_{\mathfrak{q}}$  since it is the kernel of the surjective homomorphism  $\pi_{\mathfrak{q}} : D_{\mathfrak{q}} \rightarrow \text{Gal}(l/k)$ , so  $E/K$  is normal, hence Galois (it must be separable since  $L/K$  is), and it follows that  $\text{Gal}(E/K) \simeq D_{\mathfrak{q}}/I_{\mathfrak{q}} \simeq \text{Gal}(l/k)$ .  $\square$

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